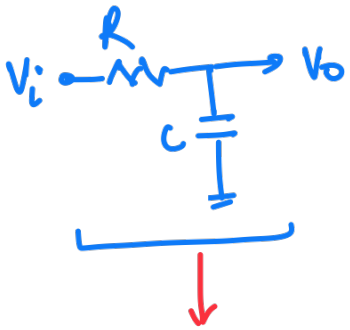
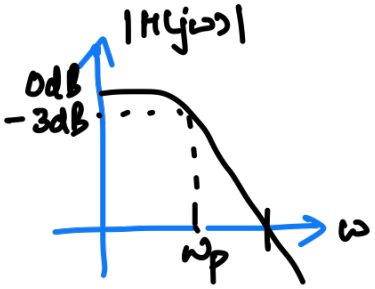
Lecture #28

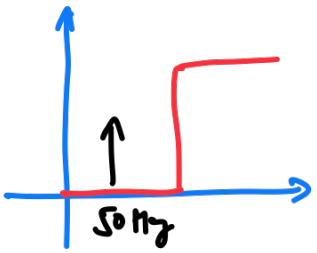
Low Pass filter



$$H(s) = \frac{1}{1+sRC}$$

$$\omega_p = \frac{1}{RC}$$

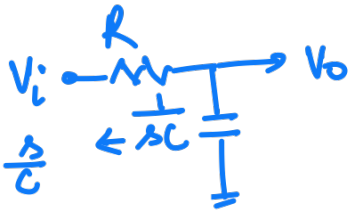




LPF	HPF
$s = 0$	$s = \infty$
$(j\omega)$	
$s = \infty$	$s = 0$

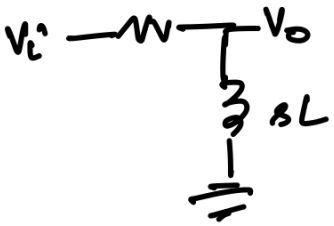
$$s \to \frac{1}{s}$$

$$H(s) = \frac{1}{sRC+1} \xrightarrow{s \to \frac{1}{s}} \frac{1}{\frac{RC}{s}+1} = \frac{sL}{R+sL}$$



$L_A = \frac{s}{C}$

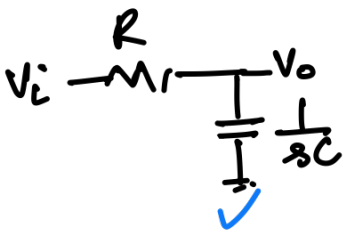
$L = \frac{1}{C}$

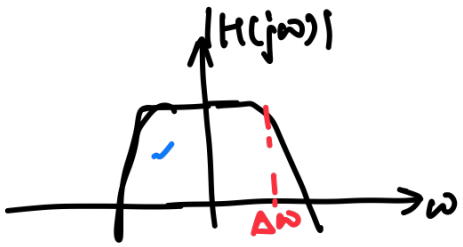


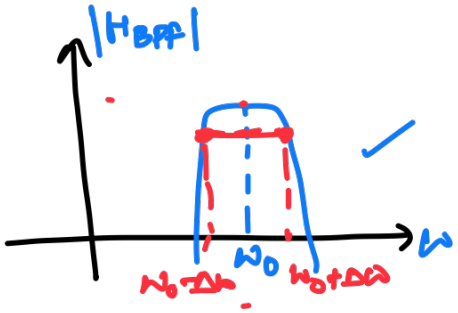
$$\frac{V_o}{V_i} = \frac{sL}{R+sL}$$

$$H(s) = \frac{sL}{R+sL}$$

$$\angle H = 90^\circ - \tan^{-1}\left(\frac{\omega L}{R}\right)$$







$s = 0$

$s = \infty$

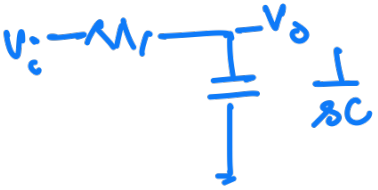
$s = j\omega_0$

$s = 0, \infty$

$s \longrightarrow \frac{s}{\omega_0} + \frac{\omega_0}{s} = \left(\frac{s^2+\omega_0^2}{\omega_0 s}\right) = 0$

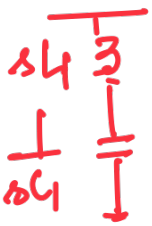
$s = \pm j\omega_0$

$H(s) = \frac{1}{1+s/\omega_p} \xrightarrow{s \to \frac{s}{\omega_0}+\frac{\omega_0}{s}} H(s) = \frac{1}{1+\frac{1}{\omega_p}\left(\frac{s}{\omega_0}+\frac{\omega_0}{s}\right)}$



$\frac{1}{sC} \to \frac{1}{\left(\frac{s}{\omega_0}+\frac{\omega_0}{s}\right)C}$

$Z_L = \frac{1}{s\left(\frac{C}{\omega_0}\right) + \frac{\omega_0 C}{s}}$



$Z_L = sL_1 + \frac{1}{sC_1}$



$Z_L = \frac{1}{\frac{1}{sL_1}+sC_1}$

$L_1 = \frac{1}{\omega_0 C}$, $C_1 = \frac{C}{\omega_0}$

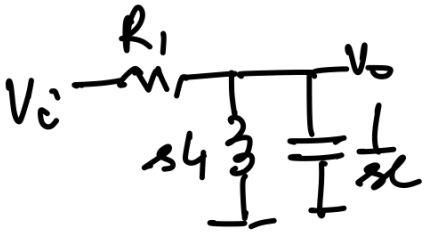
$H(s) = \frac{\omega_p \omega_0 s}{\omega_p \omega_0 s + s^2 + \omega_0^2}$

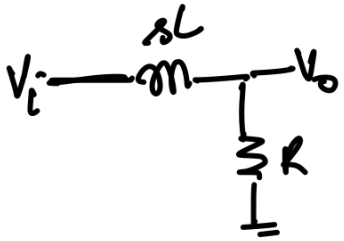
$= \frac{\omega_p \omega_0 s}{s^2 + \omega_p \omega_0 s + \omega_0^2}$

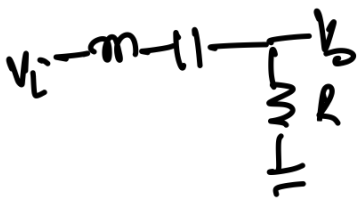
$|H(j\omega)|_{\omega=0} = 0$

$|H(j\omega)|_{\omega=\infty} = \frac{1}{\omega} = 0$

$|H(j\omega)|_{\omega=\omega_0} = \left|\frac{j\omega_0^2 \omega_p}{\omega_p \omega_0 \cdot j\omega_0}\right| = 1$



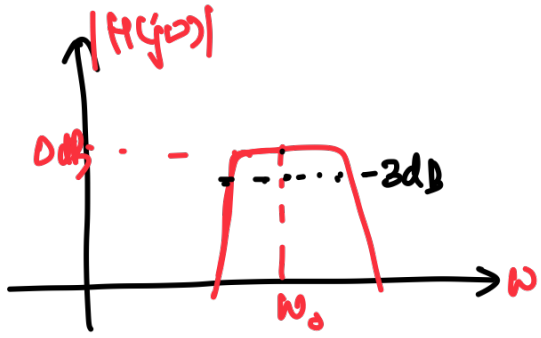




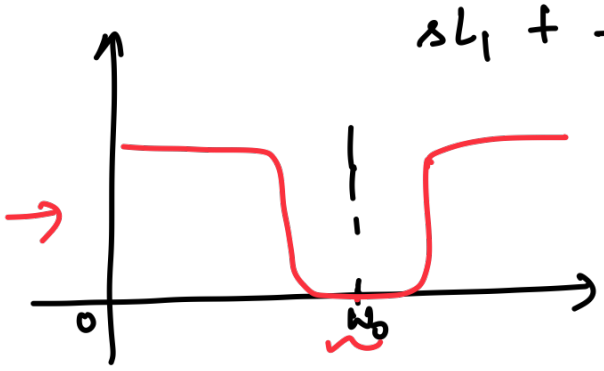
$$sL \rightarrow \left(\frac{s}{\omega_0} + \frac{\omega_0}{s}\right) L = s\frac{L}{\omega_0} + \frac{\omega_0 L}{s}$$

$$s\left(\frac{L}{\omega_0}\right) + \frac{1}{s\left(\frac{1}{\omega_0 L}\right)}$$

$$sL_1 + \frac{1}{sC_1}$$

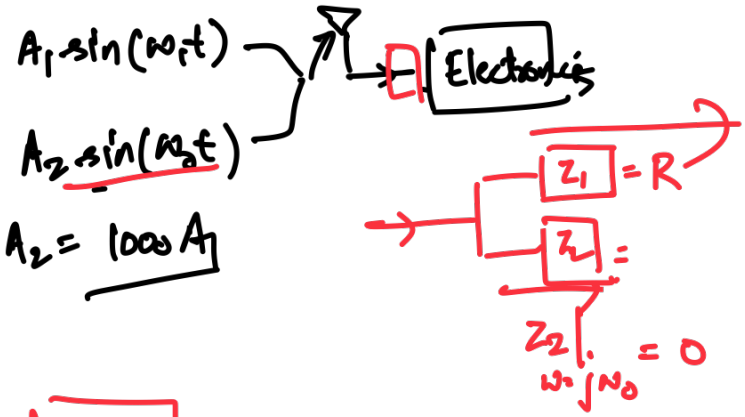


Band pass filter



Band stop filter

IEM: Intentional Electromagnetic Interference



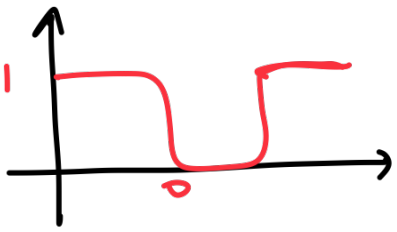


$$H_{BPF}(s) = \frac{s\omega_p/Q_p}{\frac{s^2}{\omega_p^2} + \frac{s}{\omega_p Q_p} + 1} \longrightarrow H_{BSF}$$

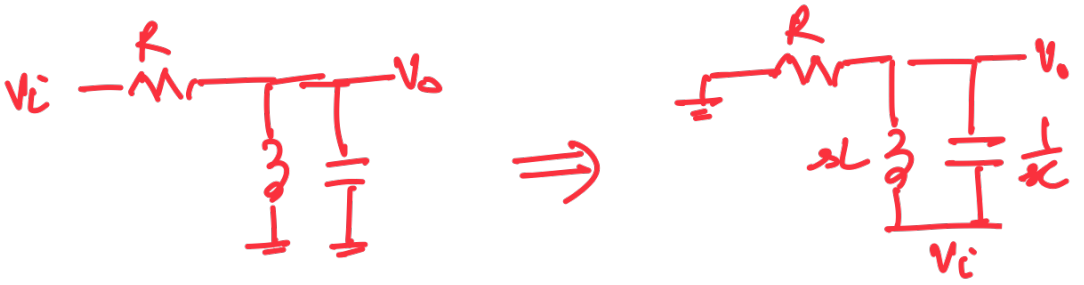
$s = 0, \infty$



$s = j\omega_0$

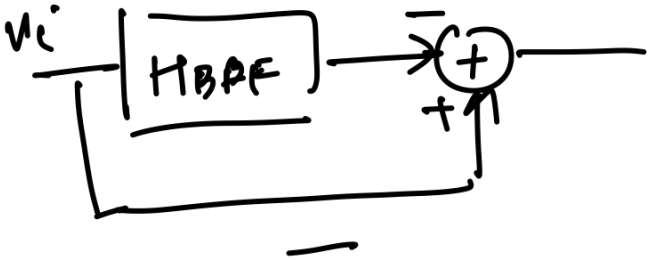


$$H_{BSF} = 1 - H_{BPF}(s)$$



$$\frac{V_o}{V_i} = \frac{R}{R + \frac{sL}{1+s^2LC}}$$

$$= \frac{1+s^2LC}{1+s^2LC+sL/R}$$

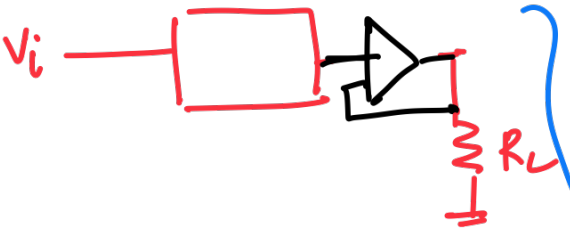


All pass filter

$$H(j\omega) = \frac{1+s/\omega_0}{1+s/\omega_0}$$

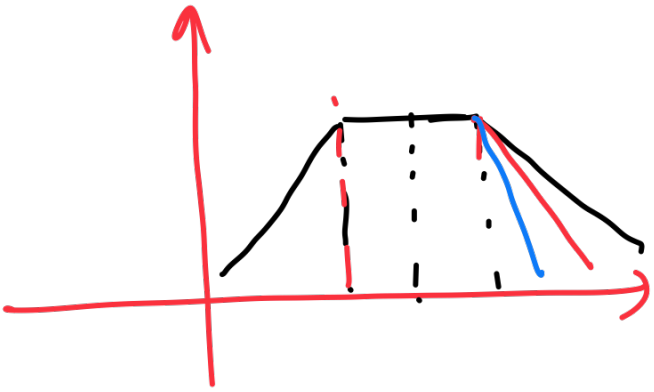


$$\angle H = \tan^{-1}\left(\frac{\omega}{\omega_0}\right) + \tan^{-1}\left(\frac{\omega}{\omega_0}\right)$$

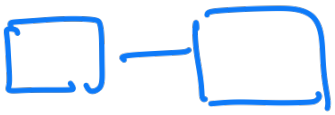


- Filtering with gain
- Inductor

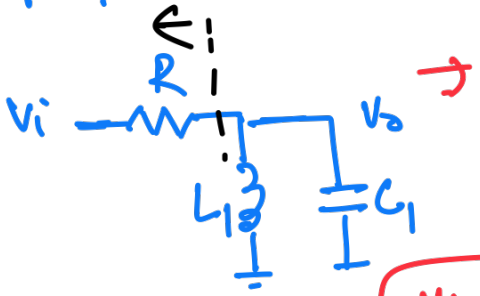
– Active filters.



$$H(s) = \frac{s/\omega_p Q_p}{\frac{s^2}{\omega_p^2} + \frac{s}{\omega_p Q_p} + 1}$$

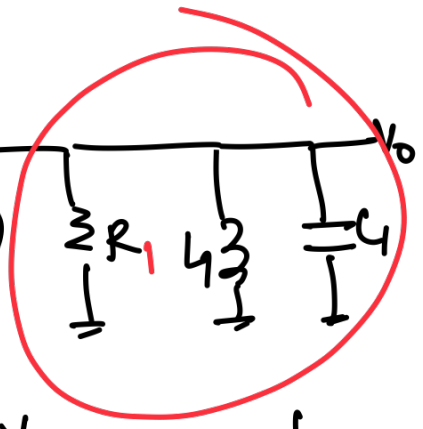


– Opamp Active-RL filters



$I = \frac{V_i}{R}$

$V_i \rightarrow I = \frac{V_i}{R_0}$



$$\left.\frac{V_o}{I}\right|_{s=j\omega} = R$$

$$\omega = \frac{1}{\sqrt{LC}}$$

$$\frac{V_o}{I} = \frac{1}{\frac{1}{R} + \frac{1}{sL_1} + sC_1}$$

$$= \frac{sL_1 R}{sL_1 + R + s^2 L_1 C_1 R}$$

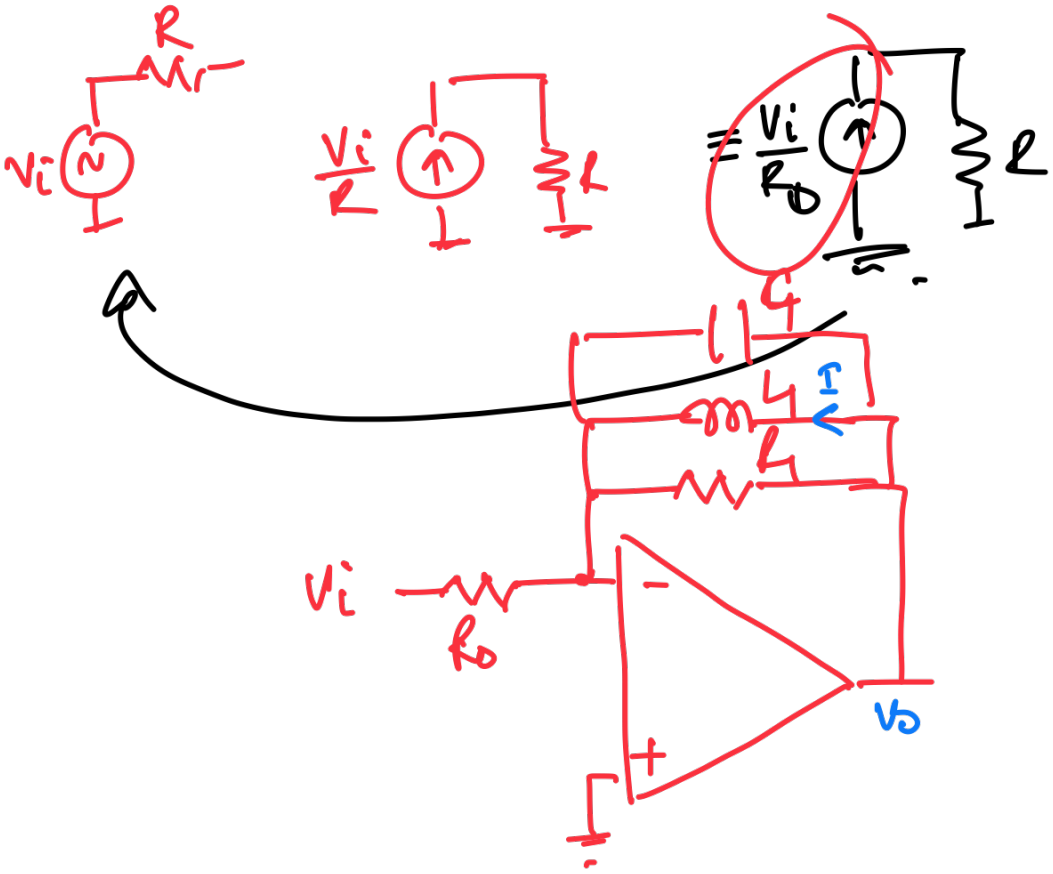
$$= \frac{sL_1}{s^2 L_1 C_1 + sL_1/R + 1}$$

$$\left.\frac{V_o}{I}\right| = \frac{sL_1}{s^2LC + \frac{sL_1}{R_1} + 1}$$

$$V_o = \frac{sL_1}{s^2LC + \frac{sL_1}{R_1} + 1} \times \frac{V_i}{R_0}$$

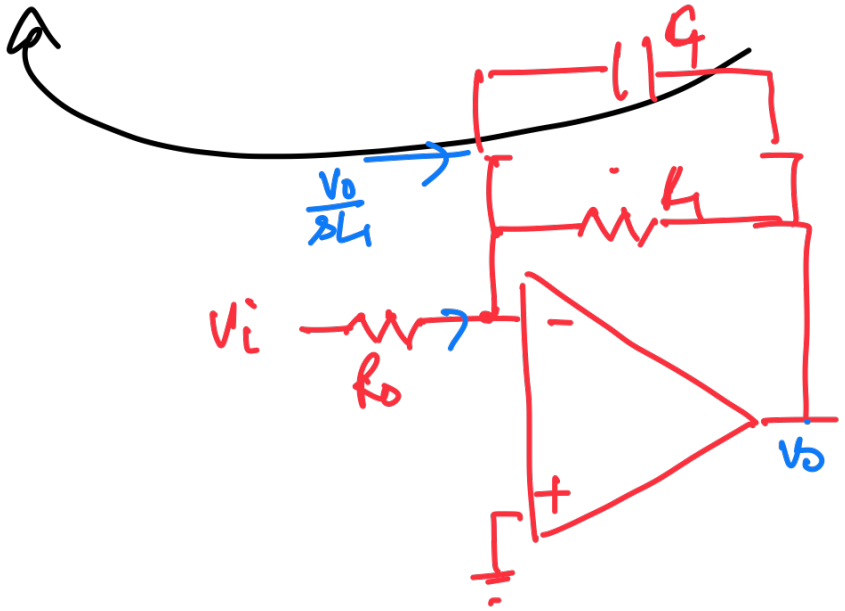
$$\left.\frac{V_o}{V_i}\right| = \frac{R_1}{R_0}$$

$$j\omega = j\frac{1}{\sqrt{LC}}$$



$$I_L = \frac{V_o}{sL_1}$$

- I_L current is integrated form of voltage V_o


C_f
$\frac{V_0}{sL_1}$
R_f
V_i
R_0
V_0